Project Plan: Assignment 1

CONSULTING PROJECT: CUSTOM GPT TO SUPPORT DISTRIBUTIVE, REGENRATIVE AND CIRCULAR DESIGN AT NIKE

MODULE: PROJECT MANAGEMENT FOR ENGINEERS MSIN0179

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Executive Summary

As part of a sustainability initiative, Nike, a global leader in sportswear manufacturing, partnered with the AI consultancy firm; Regenerative Consulting (ReCon). ReCon is developing a custom GPT model tailored to Nike's needs, facilitating the implementation of a circular economy and restoring natural resources. This collaboration aims to enhance Nike's design and R&D teams' capabilities in adopting innovative practices.

Nike has set ambitious sustainability targets, including a 65% (dunlop, 2017) reduction in scope 1 and 2 emissions, and a 30% reduction in scope 3' (Gregory, 2024) emissions by 2030, with the final goal of achieving net-zero emissions by 2050. These objectives must be met while maintaining a good product cost-quality ratio and retaining Nike's 83% customer loyalty rate.

The project plan outlines an 86-week timeline structured into four phases, from initiation to closure, which take place in seven stages. Throughout the project, several key milestones will be defined to ensure the project is going as planned with respect to its objectives, costs, and resources. This is done by collaborating with the identified stakeholders through regular meetings and workshops. Overall, the milestones will serve as indicators of project progression and success.

The ReCon team consists of 21 professionals with expertise in project management, data science, UX/UI design, sustainability analysis, logistics, accounting, and Covid compliance. Each member plays a critical role in managing budgeting, logistics, and documentation to ensure the delivery of a high-quality custom GPT solution in the expected time.

Introduction

Ranking first in the world's top 10 sportwear brand, Nike is an American athletic corporation, with headquarters based in Beaverton, Oregon, United States of America (USA). From producing footwear to manufacturing jerseys for the biggest sports teams internationally, Nike's revenues have reached $46.71 billion in 2023 (Chapman, 2023).

Section 1: Objectives

1.1 Business Need

The purpose of this project is to satisfy Nike's most recent business need, which involves implementing a custom Generative Pre-trained Transformer (GPT) tool to transition to distributive, regenerative, and circular business models. The aim is to help Nike reach its goal of reducing its environmental footprint by meeting the company's set Science Based Targets, which include an absolute reduction of scope 1 and 2 emissions by 63% and scope 3 emissions by 30%, by 2030; then reach net zero by 2050. (Nike, n.d.) (Anon., 2024). The goal is to achieve the latter by ensuring the product cost and quality is unchanged, while maintaining customer needs and satisfaction.

1.2 Main Desired Outputs

The GPT model's purpose will be to assist Nike in product development in five core areas:

Core Area	How the GPT will help	Execution and Nike's sustainability goals
Idea generation	Generating innovative and sustainable product concepts	Create new apparel which align with Nike design standards. Increase the use of environmentally preferred materials to 50%. Key materials include polyester, cotton, leather and rubber. (Nike, n.d.)
Material selection	Help designers identify and choose sustainable and regenerative materials	
Lifecycle analysis	Support the analysis of product life cycles, ensuring products are designed for reusability, repairability, and recyclability	Divert 100% of waste from landfills in Nike's extended supply chain, with at least 80% being recycled into Nike products or other goods. (Nike, n.d.)
Supply chain optimization	Help design more equitable and transparent supply	
Regenerative processes	Assist in the creation of products that contribute to the regeneration of ecosystems and reduce environmental footprints	Reduce greenhouse gas emissions by 0.5 million imperial tons less. Reduce freshwater usage per kg by 25% in textile dyeing and finishing. (Nike, n.d.)

Table 1. Project's main desired outputs based on 5 core areas

1.3 Outcomes of the Project

The outcome of the project will include fine-tuning an existing GPT to aid Nike in achieving its goals of developing products that promote equitable access to resources and value (distributive design), creating systems that restore natural resources and ecosystems (regenerative processes), designing products that minimize waste and are either biodegradable or fully recyclable (circular economy), all while adhering to

Nike's current internal design standards, and producing designs compatible with the company's manufacturing facilities and design systems. These goals should comply with external legislation and standards, thus, the GPT must ensure that what it produces is pertinent regulations and industry design standards.

Section 2: Success Criteria

The project can be deemed as successful if the objectives stated in the previous section can be achieved while maintaining a balanced project management triangle, which includes scope, cost, time on the vertices, and quality in the middle (Asana, 2024). This is summarized in Table 2:

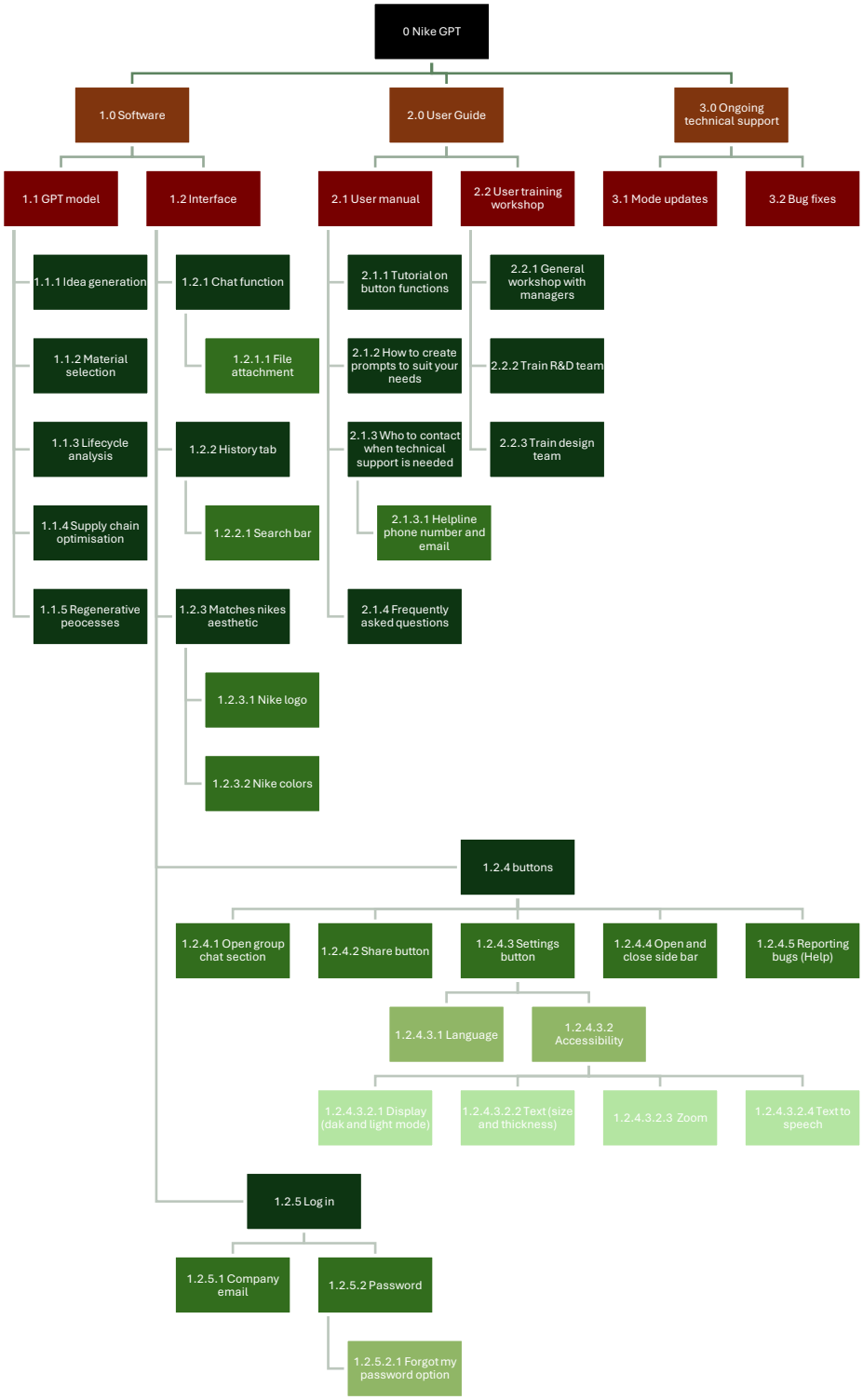
Table 2. Project's success definitions based on different criteria

Criteria	Success Definition
Scope: objectives, deliverables, boundaries, requirements	The GPT model adheres to Nike's goal of substantially reducing its environmental footprint as discussed in Section 1. This criterion will involve changes in apparel and packaging material, and using environmental-friendly manufacturing tools, while simultaneously adhering to Nike's sustainability industry standards (Nike, n.d.): • Manufacturing: Manufacturing Restricted Substances List • Chemistry: Restricted Substances List (RSL), ZDHC Wastewater Guidelines, Nanotechnology Material Guidelines, Odor Management, and Antimicrobial & Scented Materials Policy • Product: Animal Skins Policy, Amazon Biome Leather Sourcing Policy, Conflict Minerals Policy and Packaging Restricted Substances List
Time: schedule, milestones, deadlines, duration	The GPT is designed and delivered in a span of 86 weeks, following the timeline and milestones elaborated in Section 5. This report follows the assumption that the brief was handed out in August 2024. The project was therefore assumed to have started on September 23, 2024, and is expected to be completed by May 11, 2026.
Quality: standards, compliance, performance, assurance	The quality of the new designs produced by the GPT must be unchanged from Nike's current designs, or better. The project implementation must also be completed with accordance to Nike's human rights and labor industry standards (Nike, n.d.), which includes adhering to: • UN Guiding Principles on Business and Human Rights • OECD Guidelines for Multinational Enterprises • Respect for Human Rights in Nike's Supply Chain • Nike's Code Leadership Standards • Freedom of Association • Discrimination and Gender Equity • Compensation and Working Hours • Commitment to a Healthy and Safe Working Environment • Monitoring and Remediation

	• Effective Grievance Processes and Worker Management Communication
Budget and Cost: budget, expenses, funding, resources	The budget of the project remains within its estimated to £5,991,270. This amount takes into consideration everything from materials, to logistics, to launching the GPT model, until project closure. This is further elaborated in Section 7. Costs account for people (resources). This is done by appropriately choosing a team of 21 experts. This is further elaborated in Section 6.
Stakeholder Satisfaction: expectations, engagement, feedback, approval	Every stakeholder in the project was taken into consideration. Correct analysis was conducted with regards to their power and influence in implementing the GPT. This is further elaborated in Section 8.
Risk Management: identification, assessment, mitigation, control	Risks are identified correctly through a risk analysis and risk profiling. These are further elaborated in Section 9.
Value: benefits, impact, return, efficiency	In this case, the value that is intended to be created is related to making the organization more sustainable and environmental-friendly, while simultaneously creating profit, and tackling the ethical concerns raised by multiple stakeholders.

Section 3: Product Breakdown Structure (PBS) of deliverables

3.1 PBS schematic



3.2 PBS discussion

From the schematic above, the main product can be split into three deliverables: developing the software, delivering the user guide and ongoing technical support. The software is broken down into two main components: the GPT model and the interface where the GPT model is designed to meet the objectives mentioned in section 1.3. The interface is made accessible and easy to use, while also ensuring privacy by including a log in section.

The user guide deliverable, more specifically the training workshops, will equip the design team, and the research and development team with the skills to use the GPT effectively. These workshops aim to guide both teams in exploring and adopting environmentally preferred materials. Designers will find new ways in which they can make diverse products that can be reused, or repurposed. Simultaneously the R&D team will gain insights into developing new, and responsive disposal methods contributing to an improved product lifecycle. These workshops will help the teams use the GPT to meet all their aims.

Furthermore, Nike will receive a comprehensive user manual, providing guidance on how to use the GPT interface, who to contact when in need of urgent technical support and a frequently asked questions section, where common questions will be answered to save time when attempting to solve an issue.

Once this GPT has been presented to Nike, ReCon's IT team will provide continuous technical support. This is because the GPT software will not be shared to Nike's IT team in accordance with ReCon's terms and conditions on information distribution. If any bugs are reported through the help button or helpline, ReCon's IT team will work to fix the issue within 24 hours. Furthermore, to ensure the efficiency of the GPT model monthly checks and surveys will be sent out to ensure the required objectives are met, and any required updates can be made according to the responses.

Section 4: Work Breakdown Structure (WBS)

4.1 WBS Schematic

Life Cycle Phase	Level 1	Level 2	Level 3
Initiation	1.0 Project Launch	1.1 Logistics	1.1.1 Book Accommodations
			1.1.2 Arrange meals, laundry and telecommunication services
			1.1.3. Book flights and transportation
		1.2 Resource Plan	1.2.1 Discuss how resources will be deployed effectively on the project.
			1.2.2 Prepare a Responsibility Matrix.
			1.2.3 Demonstrate project feasibility with available resources
		1.3 Costing information and Cost Breakdown Structure	1.3.1 Cost breakdown structure
			1.3.2 Cost estimations
		1.4 Stakeholder Management: Discuss strategies for	1.4.1 Establish stakeholder engagement matrix
			1.4.2 Stakeholder communication plan
		1.5 Risk Management	1.5.1 Complete risk register template and describe identified risks for the project
			1.5.2 Identify control strategies and contingency plans related to the identified risks.
			1.5.3 Evaluate risks with P/I matrix.
Planning	2.0 Discovery	2.1 Needs Assessment (Anon., n.d.)	2.1.1 Develop questions to ask Nike
			2.1.2 Meetings with Nike's design and R&D teams to gather requirements
			2.1.3 Record responses and gather data
		2.2 Sustainability Goals Alignment	2.2.1 Meet with Nike's sustainability team to understand specific goals
			2.2.2 Discuss distributive, regenerative, and circular design which align with Nike's industry standards
		2.3 Current workflow analysis	2.3.1 Collect data on how Nike currently operates
			2.3.2 Use data to identify where GPT can provide most value
		2.4 AI use case brainstorming	2.4.1 Conduct workshops to brainstorm potential AI use cases that align with Nike's sustainability goals and design processes.
			2.4.2 Write a report collecting all the use cases
		2.5 Data Review: Review Nike's Data Repositories to aid in the fine tuning of the GPT model	
		2.6 Gap Analysis: Identify gaps in Nike's existing data, and areas that may serve as	

		obstacles into implementing the GPT successfully	
		2.7 Definition phase completion	2.7.1 Finalize deliverables list and create requirements document
			2.7.2 Assign roles
	3.0 Data preparation	3.1 Data collection	3.1.1 Collect relevant data from Nike's product design and R&D teams
			3.1.2 Collect sustainability reports
		3.2 Data Structuring: Clean and structure data for finetuning the GPT model (Tableau, n.d.)	3.2.1 Remove duplicate records, correct formatting, standardize data
			3.2.2 Data architecture design
			3.2.3 Document and report data
		3.3 Data Annotation: Label the data for specific tasks (including material selection and lifecycle analysis) (Abramov, 2023)	3.3.1 Choose annotation tool
			3.3.2 Define annotation guidelines
			3.3.3 Train the annotators
			3.3.4 Conduct a quality check
			3.3.5 Iterate and improve
		3.4 Data Validation: Ensure that the gathered and prepared data is accurate and ready for training (Tibco, n.d.)	3.4.1 Confirm data set contains all required data
			3.4.2 Validate data against defined rules
			3.4.3 Debug errors
		3.5 Data Report: Create data readiness report	
Execution	4.0 Fine-Tuning	4.1 Model Setup: Prepare the GPT model for fine-tuning based on the defined use cases	
		4.2 Fine-tuning Collected Data	4.2.1 Choose pretrained model and the annotated data set
			4.2.2 Load the annotated data
			4.2.3 Use a tokenizer to process the data
			4.2.4 Initialize base model
			4.2.5 Train the model
			4.2.6 Evaluate model performance
		4.3 Model Optimization: Optimize model performance for sustainability-specific tasks	4.3.1 Model pruning
			4.3.2 Data set distillation
			4.3.3 Regularization
		4.4 Model Performance and Accuracy: Internally test the fine-tuned model for performance and accuracy (Anon., 2024)	4.4.1 Functional
			4.4.2 Non-functional
		4.5 Model Feedback: Incorporate feedback from	4.5.1 Gather feedback
			4.5.2 Structure the gathered responses

		Nike stakeholders into the model training	4.5.3 Identify priorities that align mostly with end goals
			4.5.4 Implement change
	5.0 Testing and validation	5.1 Real world testing (Portal, 2024)	5.1.1 Determine methodology
			5.1.2 Describe expected outcomes
			5.1.3 Create test environment
			5.1.4 Test in the created environment
		5.2 Model Performance Evaluation: Evaluate the model against predefined metrics for accuracy and usability	
		5.3 Adjustments and Improvements	5.3.1 Collect feedback from Nike's design team
			5.3.2 Identify gaps
			5.3.3 Fine-tune and make relevant adjustments
		5.4 Final Validation: Final validation of performance and prepare for deployment	
	6.0 Initial Deployment and Support	6.1 Model Deployment: Deployment of the custom GPT into Nike's systems	
		6.2 User Training	6.2.1 Design presentations and write up user manuals
			6.2.2 General workshops with managers
			6.2.3 Train Nike's R&D team
			6.2.4 Train Nike's design team
			6.2.5 Hand out the user manuals
		6.3 Ongoing Support Set-up: Establish an ongoing support infrastructure to monitor the model post deployment	6.3.1 Train support engineers
			6.3.2 Provide Nike with communication forms
		6.4 Post-Launch Monitoring	6.4.1 Collect feedback via surveys
			6.4.2 Periodic model updates
			6.4.3 Monitoring and analytics report
Closure	7.0 Project Closure	7.1 Project closure documentation	
		7.2 Post-mortem meetings	

Figure 3. Project's Work Breakdown Structure

4.2 WBS Discussion

The table above depicts the Work Breakdown Structure which will be used to implement this project. As the backbone of the project, the WBS decomposes the total scope of work that will be completed. This includes meeting the project objectives and completing the deliverables, while abiding by its success criteria. The WBS is decomposed in a hierarchal way: Level 1 is the highest level in the table and describes the 7 stages that the project will follow – logistics, discovery, data preparation, fine-tuning, testing and validation, initial deployment and support, and project closure. These stages are divided into tasks, shown by Level 2, and further divided into subtasks/deliverables, shown by Level 3.

The WBS combines multiple structures. It is based on a blend of work-based and product-based approaches (i.e. organized around both the activities and the deliverables that are required to complete the project). It also follows the phases of a Life Cycle – initiation, planning, execution and closure – to track the project's progression over time. The WBS was elaborated using a Hybrid Life Cycle, combining both iterative and linear methods. The linear approach is seen in both the broad scale, where each life cycle phase must be completed before moving onto the next, and in a sub-task scale, such as in Task 3.0, where each sub task must be completed in a sequential way (i.e. the data annotation cannot be done before the data has been structured). The iterative approach accounts for the flexible parts of the project: an example of its application is in task 5.0, where the model is evaluated and adjusted accordingly in a feedback loop.

Overall, the structure of the WBS is done such that it ensures fixed requirements and end-products, all while allowing some flexibility for innovation and improvement wherever is needed.

Section 5: Schedule

5.1 Milestone Schedule

ID	Name	Begin date	End date	Duration
0	Phase 1: Initiation	2024-09-23	2024-10-04	10
1	MILESTONE 1: Project launched	2024-10-07	2024-10-07	0
2	Phase 2: Planning	2024-10-07	2025-05-06	152
3	MILESTONE 2: Data collection completed	2025-05-06	2025-05-06	0
4	Phase 3: Execution	2025-05-07	2026-05-01	258
5	MILESTONE 3: Finetuning and testing GPT ...	2026-01-29	2026-01-29	0
6	MILESTONE 4: GPT deployed	2026-02-13	2026-02-13	0
7	MILESTONE 5: Users trained	2026-02-27	2026-02-27	0
8	Phase 4: Closure	2026-05-05	2026-05-11	5
9	MILESTONE 6: End of project	2026-05-11	2026-05-11	0

Figure 4. Project's Milestone Schedule

The schedule above depicts the 4 phases of the Life Cycle structure of the WBS, along with the key milestones which serve as an indicator of the project progress. Each milestone is aimed to be completed by the end date mentioned in the figure above. The duration of each phase is also visible. It should be noted that these durations include the public/national holidays of both the state of Oregon in the USA and the UK. This is further discussed in section 5.2.

5.2 Detailed schedule discussion

A high-level schedule has also been provided on a separate PDF. The latter is in the form of a Gantt Chart (created using the Gantt Project Software), depicting the detailed start date and end date of every task and sub-task established in the WBS. The project was set to begin on the September 23, 2024, and end on the May 11, 2026. These dates include public holidays in Oregon, USA (Anon., 2024), such as Martin Luther King, Jr.'s Birthday (January 20, 2025), Memorial Day (May 26, 2025) or Veterans Day (November 11, 2025), as well as public holidays in the UK (Anon., 2024), such as Boxing Day (December 26, 2025) or Easter Monday (April 21, 2026). The Gantt chart also includes which tasks depend on each other (predecessors).

Employees are expected to work from 9:00 AM to 17:00 PM, with a one-hour lunch break from 12:00 PM to 13:00 PM, from Monday to Friday. The schedule is divided between tasks that overlap each other and tasks which are sequential to each other, in four main phases:

Phase 1: Initiation

The initiation phase includes the project launch, which considers all the pre-requisites for the project to begin on Nike and ReCon's behaves. Starting on September 23, 2024, and ending on October 10, 2024, its total duration is 10 working days. This phase takes into consideration all the logistics for traveling and accommodation. Two days were taken into consideration for traveling, one for the trip and one for the jetlag recovery. This phase also accounts the resource plan, costing information, stakeholder and risk managements. During this phase, many tasks overlap each other as their corresponding subtasks are either short in working days or involve multiple resources of the same role (e.g. tasks 1.3 and 1.4 are done simultaneously because there are two project managers on shift). This phase only has one milestone, which involves successfully launching the project.

Phase 2: Planning

The planning phase includes the discovery and data preparation stages. This phase starts on October 7, 2024, and ends on May 6, 2025: the total duration is 152 working days. This phase also has a mix of tasks which overlap each other and tasks which are sequential to each other. It also only has one milestone, which involves ReCon's successful data collection from Nike.

Phase 3: Execution

The execution phase includes fine tuning, testing and validation, and initial deployment and support stages. Starting on May 7, 2025, and ending on May 1, 2026, its total duration is 258 days, making it the longest phase in the schedule. The reason for this is that most tasks in this phase cannot overlap each other, rather they are sequential to each other. For instance, the improvements cannot be made unless the model has been tested, and the testing cannot be made if the model has not been optimized, etc. This phase has three milestones: the first being fine-tuning and testing the GPT model, the second being the GPT deployment, and the third being the user training.

Phase 4: Closure

The final phase of the project involves the project closure. This phase is the shortest as it serves as a project finalization stage. It begins on May 5, 2026, and ends May 11, 2026: the total duration is 5 days. This phase only has one milestone, the final milestone of the project, which involves successfully ending the project.

Additional Notes

Further factors that were taken into consideration when establishing the project schedule need to be highlighted. These assumptions are:

1. No delays have been taken into consideration directly on this schedule, only public holidays and weekends. The reason for this is because ReCon has purposely recruited multiple people per role, to ensure that absences and delays can always be covered in the case where they are needed.
2. If one individual is required to perform multiple tasks consecutively, a second person in the same role is considered to take over, following an alternating format. This ensures that no single person is responsible for an extended period; instead, they can handle a task and then take a "break" while the second person steps in.
3. If a person is consulted for a task, they can also be responsible for another and do both in parallel. This is relevant for the Responsibility Matrix in Section 6.1.
4. Approval and informing are done remotely, there is no need to be present on-site for those tasks. This is relevant for the Responsibility Matrix in Section 6.1.
5. The project manager is expected to be on-site once a week, for a weekly meeting that ensures the project is going as planned.

Section 6: Resource Plan

6.1 Responsibility Matrix

The responsibility matrix below, provides a detailed overview on the specific tasks within this project, along with who is involved in each of these tasks. The table is filled out according to the RACI acronyms (Matthews, 2024), these are:

- R: Responsible: The person that is responsible for completing the tasks
- A: Approve: The person that approves or signs off the work
- C: Consult: The person that is consulted during the decision-making process
- I: Inform: The person will be updated on the task as it is in progress

WBS Code	Technical project director	Data science engineer	Data scientist	Natural language processing (NLP) Engineer	UX/UI Designer	Sustainability analyst	Support engineer	Project Manager	Relevant stakeholders
1.0									
1.1									
1.1.1	I	I	I	I	I	I	I	R, A	
1.1.2	I	I	I	I	I	I	I	R, A	
1.1.3	I	I	I	I	I	I	I	R, A	
1.2									
1.2.1								R, A	
1.2.2	I	I	I	I	I	I	I	R, A	
1.2.3	C							R, A	
1.3									
1.3.1								R, A	I
1.3.2								R, A	I
1.4									
1.4.1								R, A	
1.4.2								R, A	I
1.5									
1.5.1								R, A	

1.5.2								R, A	
1.5.3								R, A	
2.0									
2.1									
2.1.1		I	R			R			
2.1.2		I	R			R			
2.1.3		R	R			R		I	
2.2									
2.2.1		I	R			R			
2.2.2	C	R	I			R, C		I	
2.3									
2.3.1		I	R			I			
2.3.2	I	R	R			I			
2.4									
2.4.1	C	R	R			R			
2.4.2	C	C	R			I		I	
2.5		R	R			R			
2.6	C	R	R			R			
2.7									
2.7.1	I	C	R			C		I	
2.7.2	I	I	I	I	I	I	I	R	
3.0									
3.1									
3.1.1		I	R	I		I			
3.1.2		I	R	I		R			
3.2									
3.2.1			R	I		I			
3.2.2	R	I	I	I		I			
3.2.3	I	I	R	I		I		I	
3.3									
3.3.1	C	R	R	R		C			
3.3.2	C	R	R	R		C			
3.3.3	C	R	R	R		C			
3.3.4	C	R	R	R		R			
3.3.5	C	R	R	R		I			

3.4									
3.4.1	A	C	R	C		I		A	
3.4.2	A	C	R	R		I		A	
3.4.3		R	R	R		I			
3.5	A	I	R	I		I		A	
4.0									
4.1	I	R	I	R	C				
4.2									
4.2.1	C	R	C	R	R				
4.2.2	I	R	I	R	I				
4.2.3	I	R	I	R	I				
4.2.4	I	R	R	R	I				
4.2.5	C	R	I	R	C				
4.2.6	C	I	R	R	R			I	
4.3									
4.3.1	I	R	C	R	R				
4.3.2	I	R	C	R	R				
4.3.3	I	R	C	R	R				
4.4									
4.4.1	I	R	R	R	I				
4.4.2	I	R	R	R	I				
4.5									
4.5.1	I	I	I	I	I			R	
4.5.2		I	I	I	I			R	
4.5.3	C, A	R	R	R	R			I, A	
4.5.4	C	R	R	R	R			I	
5.0									
5.1									
5.1.1		R	R	R	R		C		
5.1.2	C	R	R	R	R		R	I	
5.1.3	C	R	C	R	C		I		
5.1.4	I	R	R	R	I		I		I
5.2		C	R	I	I		I	I	
5.3									
5.3.1		I	R	I	I		I	R	

5.3.2		C	R	C	C		C	R	
5.3.3		R	R	R	R		I		
5.4	C, A	R	R	C	C		C	A	
6.0									
6.1	A	R	C	C	C	C	I	R, A	I
6.2									
6.2.1				C		C	R	A	
6.2.2				I		I	R	R	
6.2.3				I		I	R	C, I	
6.2.4				I		I	R	C, I	
6.2.5				I		I	R	C, I	
6.3									
6.3.1		R				I	R	R	
6.3.2	I	I				I	R	C	
6.4									
6.4.1		I				I	R	I	I
6.4.2		R				R	R	I	
6.4.3		R				R	R	I	I
7.0									
7.1	I	I	I	I	I	I	I	R, A	I
7.2	I	I	I	I	I	I	I	R, A	

Table 5. Project’s Responsibility Matrix

6.2 Histogram and S-Curve

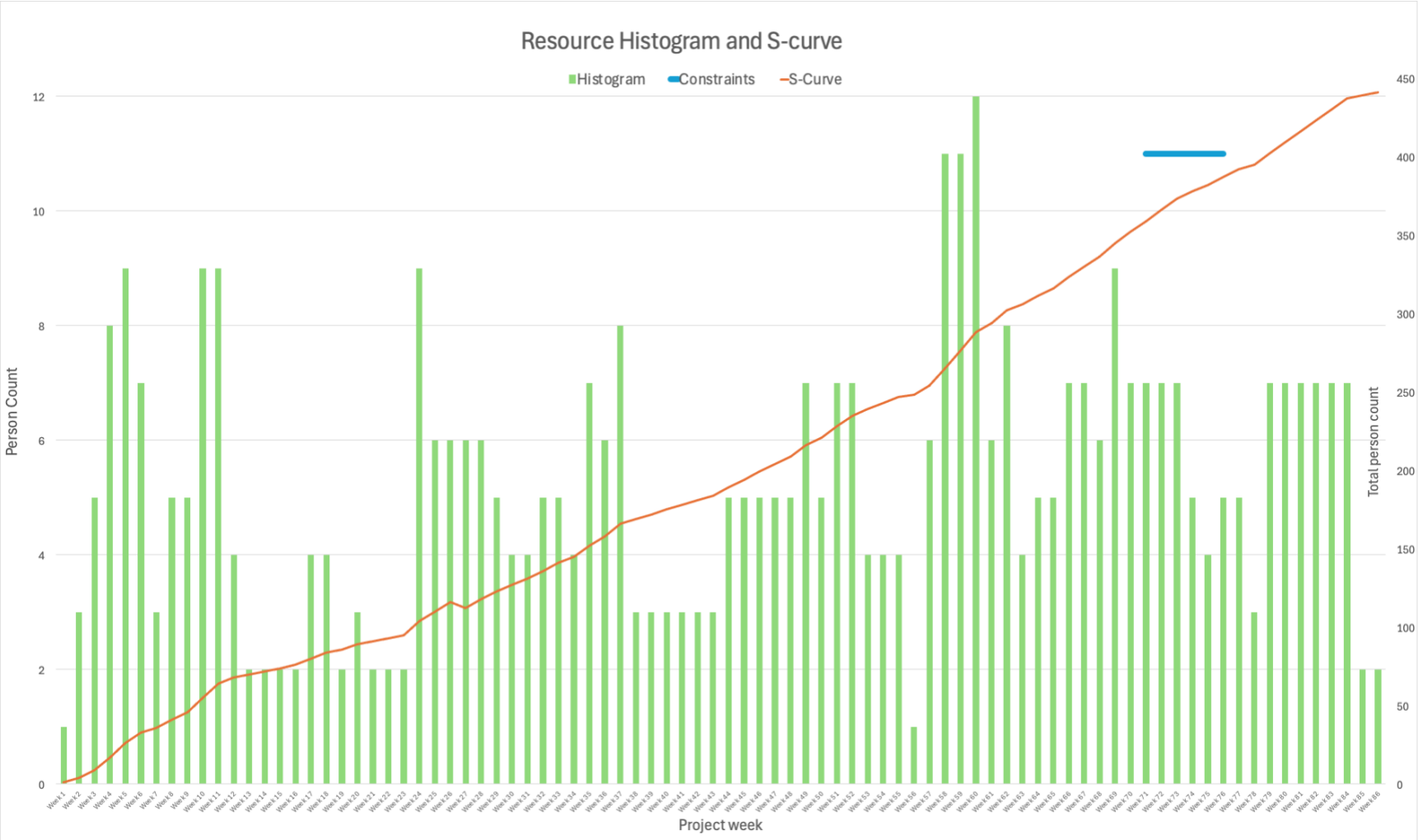


Figure 6. Project's Resource Histogram and S-curve

6.3 Histogram and S-Curve Discussion

The graph above combines a histogram and an S-curve to help understand the resources deployed throughout the project. The graph was established according to the data in the responsibility matrix, and the Gantt chart. The blue line represents a resource constraint, between weeks 71 and 76, during which only 11 members of the team will be present on the US site. This graph allows for the comparison between the project progress and plan, highlighting performance, trends and variances. This allows quick identification of issues and helps in predicting outcomes.

The histogram represents the distribution of resources (number of people working) weekly. As can be seen above this resource distribution varies according to the week. For instance, in the first week, only the project manager is responsible for the initiation phase, whereas in week 58, 11 people are expected to work.

The S-curve represents the cumulative resources over time. As can be seen at the beginning of the graph, there is a flatter gradient, this is because during the initiation phase not many people are expected to work, and most of the work will be done by the project manager. Between weeks 21 and 84 there is a steady growth in the S-curve. After week 84, the S-curve begins to plateau because during the closure phase, most people are done with their job, and only the project manager is responsible for finalizing all the required documents.

Overlapping the histogram and S-curve allows for the comparison of the two. Some resources are spread out unevenly (e.g. comparing weeks 21 to 24). In theory efficiency would be improved by distributing resources more evenly, simultaneously causing a smoother and more prominent S curve.

Section 7: Costing Information

7.1 Cost Breakdown Structure

Considering this project entails 4 phases, spread along 86 weeks, it is important to divide the costs into various categories. This ensures a clear plan is kept on what expense are expected throughout the progression of the project.

According to the briefing summary the categories are as follow:

1. Logistics Costs: logistic staff salary including technical support manager and assistant, covid compliance assistant, accountant and computer system technicians.
2. Labor Costs: Employees wages, this includes anyone who is part of the team from technical project director to support engineer.
3. Services Costs: Accommodation, Food and Telecommunication Services.
4. Travel Costs: Transportation of needed team members from ReCon in the UK to Nike in the USA, and back (this accounts for the compensation fee per person travelling to the USA).

Calculations in the following section were done for each of the above categories, in each stage of the project. These calculations are based on information gathered from the brief, paired with research and assumptions. The following calculations allow the project manager and accountant to monitor the spending, ensuring that early changes can be made when necessary to avoid going over budget.

7.2 Costing Assumptions

To successfully estimate the budget costs, assumptions needed to be made in certain areas of the project, such as:

- The exchange rate between pounds and dollars was assumed to remain constant throughout the project, where 1 Great British Pound (£) = 1.27 American Dollars ($) (Anon., 2024), according to the latest news.
- Food is covered for team members during the entire stay in the USA.
- The total length of the project is 83 weeks, this includes weekends and public holidays, however they were not included in the cost budget considering that these days are not paid.
- Installation of telecommunication services takes 4 weeks; they are therefore installed a month prior the team flying to the USA. Therefore, the overall cost of telecommunication services will equal the operating length plus 4 weeks of installation.
- The cost of flight was estimated using United Airline, a full-service American airline, in economy (to limit the fees).
- The covid compliance assistant cost is neglected as the project is lead between the 2025-2026 period.

7.3 Cost Estimations

Separate tables were made for the 4 different categories of cost that need to be estimated. These can be found below.

Logistics	Prices	Justification
Logistics Manager	$5{,}550 \times 2.1 = £11{,}655$	The logistic manager will be needed during two phases: • **Fine Tuning**: To plan and prepare all the logistics required for the presentation in the USA (approximately one month) • **Deployment & Support**: To ensure proper transportation and logistics deployment for the team throughout their stay in the USA (33 days) **Total Estimate:** 63 days = 2 month + 3/30 **= 2.1 month**
Logistic Assistant	$3{,}950 \times 2.5 = £9{,}875$	The logistics assistant will work approximately two months in phase one and can be considered on part time employment in the third phase if their services are required: • Fine-tuning phase: Full employment to assist with paperwork and accommodation preparation. • Deployment and support phase: Part-Time employment in case any logistics issues were to rise when the team is in the USA **Total Estimate: 2.5 months**

Covid Compliance Assistant	£0	Considering the project occurs in the 2025-2026 period, a covid assistant is not considered necessary for this project.
Accountant	$6{,}550 \times 19.63 =$ £136,438.5	The accountant is a central part of the project to ensure, its successful completion, and is therefore expected to work one day per week. Total Estimate: Project Begins: 23/09/2024 and ends 11/05/2026: 596 days = 1 year, 7 months, 19 days = 19 months, 19 days = 19 +19/30 = 19.63 months
Computer Systems Technician	$4{,}450 \times 19.63 =$ £85,573.3	Required for the: Initial On-Site Support: 1 week at the beginning of Stage 1 Ongoing Remote Support: Estimated about 1-2 days per month, to address any issues. The computer systems technician is required every month, however, can work remotely and therefore will not fly to the USA.
Total cost		$= £\mathbf{243,541.8}$

Table 7. Project's logistics, prices and justification (explanations)

Services	Cost	Justification
Accommodation (per person per night)	$290 = £228.35 Therefore, Total cost: $228.35 \times 11 \times 33 = £82{,}891$	Airbnb is preferable to a hotel due to cost efficiency considering the length of stay in the US. The Airbnb will cost £228.35 per person, per night.
Cost of return flights from London to Washington County, near Beaverton, Oregon, United States.	Total cost: $900 \times 11 = £9{,}900$	Choice of economy seats for cost efficiency but still booked tickets through a trusted airlines like United. £900 pounds per person with United airlines (Anon., 2024) from London Heathrow to Portland international airport.
Travel arrangement to client site	• Bus from Bedford to London: £150 • A return bus (rent of minibus for the entire team) from Airport to Airbnb: £240 • When travelling from Airbnb to the office and vice versa the team will use a public transport bus within	An Airbnb has been booked close to Nike, allowing team members to use public transport when travelling to and from the office.

	Beaverton: return tickets cost £2.80 • Overall public transport costs: $2.80 \times 5\ days\ per\ week \times 5\ weeks =$ £70 per person • $70 \times 11 =$ £770 Total cost: $150 + 240 + 770 =$ £1,160	
Telecommunication services per month	$480 = £377,95 Initial development = 10+20+15+30 = 75 days = 1 months and a half. + 1 month for installation. • Employ them for two months and a half so: $480 * 2 + 480/2 = \$1{,}200\ dollars = 944.9\ pounds$	4-week installation = 1 month Necessary to install them before setting up the model to ensure their correct functioning. The services are therefore needed during the initial deployment and support stage.
Food per person per week	$480 = £377.96 per person Team members will go to the USA for 33 days = 5 weeks. Team members going to the USA: • 1 NLP Engineers • 2 Data scientist • 2 Data science engineer • 1 Sustainability analyst • 1 Support engineer • 1 Project managers • 1 Technical project director • 1 Logistics manager • 1 Logistics assistant The accountant will work remotely Taking 11 people to the USA: $5 \times 11 \times 377.97 =$ £20,787.25	There are 11 team members flying to the USA, for 5 weeks, this is because the work that will be done in the USA does not require all 21 team members, this would be an unnecessary cost. During the stay in the USA food is covered for every individual, every day of the week. This will cost £377.96 per person per week.
Disruption Allowance	$65 \times 11\ people \times 33\ days =$ £23,595	A disruption allowance is available to all members required to work abroad for this project. It is equal to £65, per person per night.
Total cost		$= £\mathbf{139,278.15}$

Table 8. Project's services, costs and justification (explanations)

Roles	Number of Employees	Salary	Number of Working Days	Total Cost
Technical Project Director	1	£2,250	Twice a week throughout the whole project: 83 weeks in the project. Works twice a week = 166 Workdays	$2{,}250 \times 166 = £373{,}500$
Project Manager	2	£1,850	2 day/ week each So, 166 days of work each throughout the project	$166 \times 1{,}850 \times 2 = £614{,}200$
Data Science Engineer	3	£1,650	$15 + 10 + 10 + 10 + 5 + 5 + 20 + 30 + 25 + 15 + 10 + 60 + 20 + 20 + 15 + 30 + 10 + 20 + 10 + 10 + 15 + 30 = 395$	$395 \times 1{,}650 \times 3 = £1{,}955{,}250$
Data Scientists	3	£1.400	$15 + 10 + 10 + 5 + 5 + 20 + 30 + 25 + 15 + 10 + 60 + 20 + 20 + 15 + 30 + 10 + 20 + 10 + 10 = 350$	$1{,}400 \times 350 = £490{,}000$
Natural Language Processing (NLP) Engineer	2	£1,600	$20 + 10 + 10 + 20 + 10 + 30 + 15 + 20 + 60 + 10 + 15 + 25 + 30 + 20 = 295$	$295 \times 1{,}600 \times 2 = £944{,}000$
UX/UI Designer	2	£1,350	$10 + 20 + 10 + 30 + 15 + 20 + 20 + 60 + 10 = 195$	$195 \times 1{,}350 \times 2 = £526{,}500$
Sustainability Analyst	2	£1,750	$30 + 15 + 20 + 10 + 10 + 20 + 10 + 30 + 15 + 25 + 30 + 20 + 5 + 5 + 10 + 10 + 10 + 15 = 290$	$1{,}750 \times 2 \times 290 = £1{,}015{,}000$
Support Engineer	2	£1,200	$30 + 15 + 20 + 10 = 75$	$1{,}200 \times 2 \times 75 = £180{,}000$
Total cost				$= £\mathbf{5{,}608{,}45}$

Table 9. Project's number of employees and number of days they work in, along with their respective salaries

7.4 Costing conclusion

The expected total cost of the entire project will be £5,991,270. The above calculations represent a comprehensive analysis on all the expected expenses. Regular monitoring on the actual value spent compared to the expected value spent will be conducted throughout the project lifetime. This allows ReCon to remain within budget and address any issues before it is too late.

Section 8: Stakeholder Management Plan

The stakeholder management plan identifies stakeholders and analyses their influence levels (Smudde & Courtright, 2011). It is essential to meet as many stakeholders' expectations as possible to ensure successful delivery (Varvasovszky & Brugha, 2000). This involves engagement actions to be clearly understood by staff and an effective communication plan to be made and followed.

8.1 Stakeholder Engagement Matrix

The stakeholders identified are listed in Table 10 with their corresponding interest and impact level and engagement action category. Actions that need to be taken to ensure increased support and/or decreased opposition from each stakeholder is also included.

Stakeholder	Interest	Impact	Engagement Action	Actions to increase support	Actions to decrease opposition
Board of Directors	2.5/10	9.5/10	Keep Satisfied	Demonstrate ROI and alignment with strategy	Address risks proactively, ensure transparency
Local Government	2/10	7.5/10	Keep Satisfied	Highlight project benefits to community	Address concerns on ethics and compliance
Middle, design and operations managers at firm	7/10	9/10	Manage Closely	Showcase productivity improvements	Minimize workload disruptions
Researchers	8/10	3/10	Keep Informed	Provide access to data/tools for research	Address ethical and privacy concerns
Local activists	9/10	1.5/10	Keep Informed	Highlight benefits	Provide transparency on data and impacts
Development Team	8/10	8.5/10	Manage Closely	Showcase productivity improvements	Minimize workload disruptions
Contractors	7/10	7/10	Manage Closely	Share relevant use cases and benefits	Ensure minimal disruption to their processes
Nike's R&D and Design Teams	7/10	6/10	Manage Closely	Share relevant use cases and benefits	Ensure minimal disruption to their processes
Press/Media	7/10	0.5/10	Keep Informed	Highlight innovation and positive impacts	Avoid negative press through timely updates
Nike's Distributors	5/10	2.5/10	Monitor	Share benefits on demand forecasting/logistics	Clarify operational impacts early
Nike's Suppliers/ Manufacturers	3.5/10	3.5/10	Monitor	Show potential for improved efficiency	Ensure no additional workload
Nike's Customers	4/10	6/10	Keep Satisfied	Involve in beta testing, offer exclusives	Address privacy concerns and ensure clarity

Table 10. Stakeholder Engagement Matrix

8.2 Stakeholder Power/Interest Matrix

Identified stakeholders are categorized based on their level of power and interest and plotted on a Power/Interest Matrix in Table 11 to further highlight the priority of stakeholders (Walker, et al., 2008). Each category is further explained below.

1. High Power, High Interest (Manage closely): These stakeholders are very influential and important to keep satisfied. Pay close attention to their needs and feedback and engage them in the process actively.
2. High Power, Low Interest (Keep Satisfied): These stakeholders must also be satisfied, however do not bore them with too much detail and communication.
3. Low Power, High Interest (Keep Informed): Communicate and engage as needed.
4. Low Power, Low Interest (Monitor): Monitor feedback and changes, regular communication is not necessarily needed.

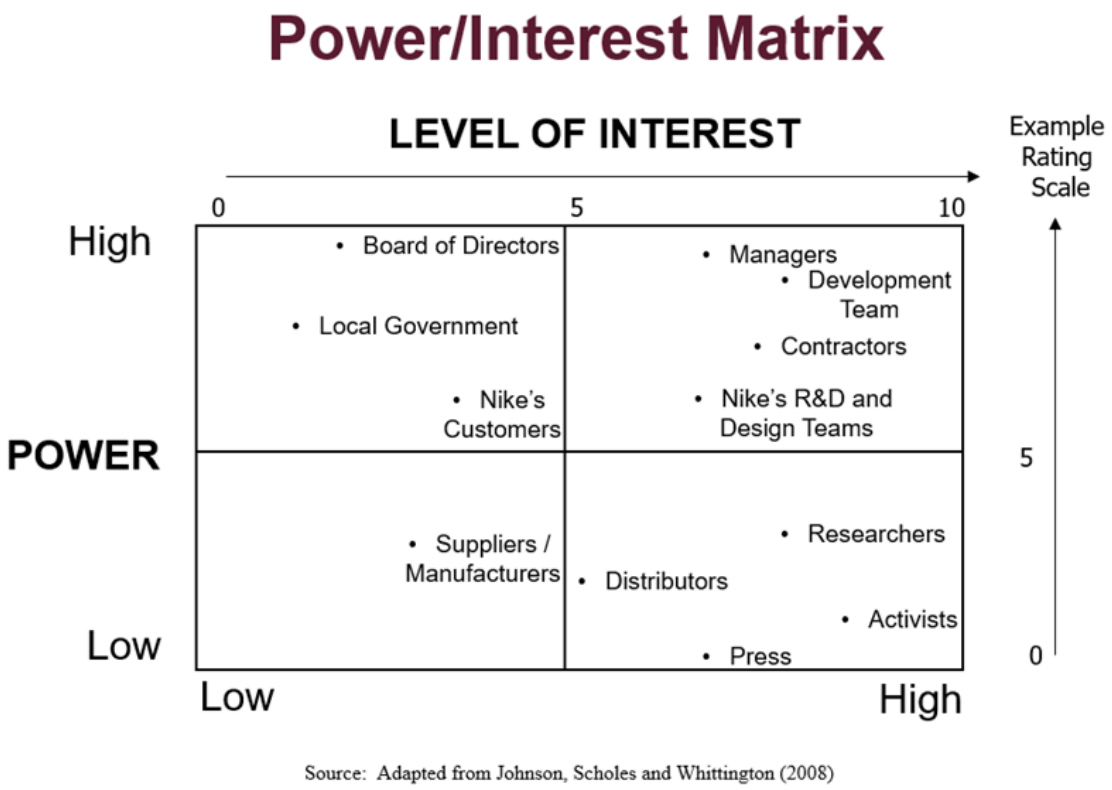


Table 11. Power/Interest Matrix of Stakeholders

8.3 Stakeholder Communication Plan

The stakeholder communication plan lays out the content, channel and frequency of communication for each stakeholder varying based on their power and interest. The desired outcome and which member of staff is responsible with the communication with each stakeholder can also be found in Table 12.

Stakeholder	Content	Channel	Frequency	Desired Outcome	Responsible
Board of Directors	Engage in decision-making; align with strategic goals	Meetings / Reports	Monthly reports, bi-weekly meetings	Full buy-in and active project sponsorship	Project Manager and Senior Leadership
Local Government	Ensure regulatory compliance, address social concerns	Reports	Monthly and open channels for queries	Supportive stance with no regulatory barriers	Legal and Compliance Team
Middle, design and operations managers at firm	Highlight operational benefits	Emails and Meetings	Bi-Weekly	Support for smooth integration into workflows	Project Manager
Researchers	Share tech insights; involve in research discussions	Workshops / Newsletter	Quarterly webinars and Targeted reports	Collaboration and endorsement of the project	Research Liaison
Local activists	Address ethical concerns; emphasize transparency	Newsletter	Webinars, and open channels for concerns	Mitigated opposition; positive advocacy	Corporate Social Responsibility (CSR) Team
Development Team	Highlight potential improvements in workflow	Meetings and emails	Daily	Support for smooth integration into workflows	Project Manager
Contractors	Project insights, minimize unnecessary details	Email	Bi-weekly	Alignment and smooth collaboration	Project Manager
Nike's R&D and Design Teams	Project insights, minimize unnecessary details	Email	Weekly	Alignment and smooth collaboration	Project Manager and Liaison
Press/Media	Control narrative; share milestones	Press releases	As needed	Positive media coverage	Public Relations Team
Nike's Distributors	Inform on changes affecting distribution	Email	As needed	Improved operational efficiency and acceptance	Supply Chain Manager
Nike's Suppliers/ Manufacturers	Minimal updates; focus on supply chain impacts	Email	As needed	Neutral stance with minimal disruption	Procurement or Supply Chain Manager
Nike's Customers	Gather feedback; involve in development	Newsletter, social media engagement and surveys	Monthly	Positive customer reception and engagement	Marketing and Customer Experience Team

Table 12. Stakeholder Communication Plan

Section 9: Risk Management

Novelty and uncertainty bring risks with it and those risks must be systematically managed to ensure the success of the project. This section follows the guidance outlined in the Association for Project Management's (APM) Project Risk Analysis and Management (PRAM) Guide (Association for Project Management, 2004). The two key strategies involved are Risk Analysis and Risk Profiling as detailed in the sub-sections below.

9.1 Risk Analysis

Risk Analysis is about identifying and analyzing risks. The process involves a brainstorming session and document reviews of project deliverables. Once risks are identified, further discussions took place to assess the probability of each risk happening and its impact on the project across three dimensions: quality, cost, and time. Following this, Treatment strategies were recommended to reduce the likelihood of risks and contingency strategies to mitigate their potential impact. Each risk is assigned to a member of staff to oversee the process. The criteria for evaluating probability and impact scores are on Table 13 adapted from (Association for Project Management, 2004) guide. The Impact score averages time, cost, and quality, helping prioritize risks in the Risk Profiling section. The risk register in table 14 brings together the identified risks and their sources, probabilities, impacts, treatment and contingency strategies and responsibility assignments.

Scale	Probability	Impact on Project		
		Time	Cost (increase)	Quality
Very Low	<5%	<1 day	<5%	Failure to meet minor deliverable specifications.
Low	5-15%	1-2 days	5-10%	Failure to meet >2 minor deliverable specifications.
Medium	15-30%	2-5 days	10-15%	Shortfall in meeting deliverable specifications.
High	30-50%	5-7 days	15-20%	Significant shortfall in meeting deliverable specifications.
Very high	50%+	7+ days	20%+	Failure to meet deliverable specifications.

Table 13 Criteria for Probability and Impact Scale Ranking System

Risk ID	Risk Source	Risk Description	Probability	Impact on Cost, Quality, Time	Risk Treatment Strategy	Contingency Strategy	Risk Owner
1	Weather	Flights delayed due to unsafe weather conditions	Medium	Cost: Medium since required to re-book accommodation and flights. Quality: Low as flights are only required during certain stages which is not affecting a lot of deliverables. Time: Low as short delays are within acceptable criteria. Overall Impact: Low	Book flexible flights with travel insurance. Arrive 1 or 2 days earlier at the destination before the project starts.	Have an emergency fund for unexpected problems that may arise during travels.	Logistics Manager
2	Absence	Short term employee absence due to minor illness	High	Cost: Very low as short term illness should not affect costs Quality: High as absence of key staff will affect quality in specialized work Time: Low as short delays are within acceptable criteria Overall Impact: Medium	Promote a clean and hygienic office culture by providing sanitizing products.	Prevent the further spread of illness in the office by encouraging affected employees to stay at home. Arrange cover while the affected employee recovers, to make sure delays are minimized.	Project Manager
3	Data privacy	Data breach results in compromising user/customer data	Medium	Cost: High as security costs will increase and there will be legal fees Quality: High as there is damage to reputation Time: Medium due to time lost investigating and fixing the issue Overall Impact: High	Implement strict security measures for everyone who has access to the data. Regularly audit systems and employees.	Prepare an incident response plan and have data insurance sorted out.	Legal and IT Security Team

4	Project scope creep	Additional features that were not planned for increases the workload	High	Cost: High as budget increases by a lot for unplanned requirements. Quality: High as overextension will cause shortfall in meeting acceptable criteria. Time: High due to the time lost by uncontrolled scope growth Overall Impact: High	Define clear scope and have a deadline to finalize requirements. Implement a change management plan.	Prioritize “must have” and “should have” requirements and features. Re-establish timeline and budget.	Project Manager
5	Resource constraints	Shortage of available key staff or budget overruns	Medium	Cost: High due to overtime expenses or having to hire additional staff Quality: Medium due to rushed deliverables Time: High as significant delays due to resource shortage. Overall Impact: High	Continually assess resource allocation to stay on track with budget and to optimize workload.	Re-establish timeline or scale back features.	Project Manager
6	Project Definition	Miscommunication when capturing requirements results in the client’s expectations to be unmet.	Low	Cost: Very high as the client may not provide the expected compensation. Quality: Very high as the deliverables does not match the client’s criteria. Time: High as there will be long delays due to fixing deliverables. Overall Impact: Very High	Adopt an agile process and feedback loops to resolve any miscommunication as they occur. Project expectations to be agreed clearly on the contract before proceeding with tasks.	Meet with legal consultants to resolve legal disputes.	Consultancy Manager
7	Technical Challenges	Difficulty integrating GPT into Nike's existing systems	Very High	Cost: Medium as there will be some increase in integration cost. Quality: High as the project might not be scalable. Time: High as troubleshooting and rework results in delays. Overall Impact: High	Consult and hire technical experts and conduct feasibility testing.	Allow for extra time and resources in the plan.	Development Team Lead

8	User Acceptance	End users may not use or trust the new system	Medium	Cost: Medium as more training Nike's staff will be needed. Quality: Low as the deliverables are still within criteria Time: Medium as feedback incorporation and trainings will cause delays Overall Impact: Medium	Conduct user testing and incorporate feedback loops.	Offer onboarding support.	Liaison for Nike's staff
9	Stakeholder Resistance	Lack of buy-in from key stakeholders	Low	Cost: Medium as some additional cost needed to engage stakeholders Quality: High as business goals are not aligned with deliverables. Time: High as scope clarification and rework causes delays. Overall Impact: High	Make sure that the stakeholders are updated regularly and engaged from the beginning of the project.	Adjust project scope to satisfy key requirements. Escalate to sponsor.	Project Manager and Sponsor
10	Regulatory Compliance	Accidental breach of AI regulations and ethical guidelines.	Very low	Cost: High as there will be legal penalties and redesign costs Quality: Very high as deliverables fail to meet ethical standards. Time: Very high as significant time lost on legal reviews and adjustments Overall Impact: Very high	Make sure to review legal requirements and align with global standards.	Adjust the model post launch to meet the regulations.	Legal and Compliance Team

Table 14 Project Risk Register

9.2 Risk Profiling

Risk Profiling focuses on the evaluation and prioritization of risks based on their probability and impact. The risks discussed in table 14 are profiled using the scoring system on table 15. The probability and Impact scores are multiplied to give a priority score for each risk as shown on table 16. Impact score follows a logarithmic scale to highlight impact of severe low probability risks.

This information is then visualized on a Probability/Impact matrix on table 17 to identify the order that they should be addressed.

Scale	Probability		Impact	
	Rating	Score	Rating	Score
Very low	Rare	1	Insignificant	1
Low	Unlikely	2	Minor	2
Medium	Possible	3	Medium	4
High	Likely	4	Major	8
Very high	Almost Certain	5	Severe	16

Table 15 Displaying the Scoring system

		Impact				
		Very Low	Low	Medium	High	Very High
Probability	Very Low	1	2	4	8	16
	Low	2	4	8	16	32
	Medium	3	6	12	24	48
	High	4	8	16	32	62
	Very high	5	10	20	40	80

Table 16 Displaying the Empty P/I Matrix Scores

		Impact				
		Very Low	Low	Medium	High	Very High
Probability	Very Low					Risk 10
	Low				Risk 9	Risk 6
	Medium		Risk 1	Risk 8	Risk 3 & 5	
	High			Risk 2	Risk 4	
	Very high				Risk 7	

Table 17 Displaying all Defined Risks on P/I Matrix

9.2.1 Probability / Impact Matrix Discussion

The P/I Matrix in table 17 and using the scoring system in table 16, the highest priority risk was identified as risk 7, and second highest priority risk was identified as risk 6.

Risk 7, namely technical challenges, has a very high probability of causing disruption to the project. This is due to inherent complexity of integrating GPT with Nike's existing systems with potential incompatibility with legacy platforms and the need for extensive customization of the model. Any resource or expertise gaps may increase the likelihood even further. Since the impact on quality and time is high as discussed in

risk register, this risk would be detrimental to the success of the project. To mitigate this risk, careful planning, iterative feasibility testing and regular communication with technical experts are to be implemented.

Risk 6, namely Project definition, refers to Nike's expectations to be unmet. The likelihood of this happening is identified as low since ReCon takes great care to capture accurate requirements through an iterative approach with feedback loops to resolve any miscommunication as they occur. However, if this was to happen, the impact on cost, time and quality would be very high. Therefore, prevention by effective communication is essential. The contract must describe the obligations of both companies clearly, to avoid legal disputes.

Section 10: Project Monitoring Strategy

To successfully lead this project, it is important to closely follow each step that contributes to it. Thus, a comprehensive project monitoring strategy was created to help project leaders keep track of the progress made in specific expected timeframes. The monitoring strategy's goal is to ensure the set objectives and milestones are met, while also allowing the board to identify potential issues early enough to implement solutions. This section covers various areas of the project, including the budget, schedule, quality monitoring, as well as the employment of resources and project summary.

10.2 Summary of Factors to Monitor

To summarize the different monitoring techniques and assess their impact on the project, it is important to first determine which factors are fundamental to the successful development of the project. Table 18 summarizes the factors that need to be monitored closely and the steps that can be done to address them:

Key Factors	Monitoring solutions	Justification
Project Progress, Data and Success	• Create a Gantt Chart and WBS. • Hold weekly meetings with project managers. • Monthly meeting with technical project director.	It is essential to divide the milestones into smaller timeframes to ensure both large and smaller scale objectives are clear. This approach allows comprehensive monitoring of progress and early implementations of solutions if issues arise.
Time Delivery	• Regularly check the progress of various project steps by following the Gantt chart. • Hold weekly meetings with project managers.	Meeting project deadlines is crucial to maintaining the client's trust, upholding the reputation of the company and avoiding potential economic repercussions

Cost	• Work closely and regularly with the accountant. • Plot an S-curve before beginning the project to have a rough estimation of the expected cost. • Conduct weekly checks on the budget spent. • EV chart	Can keep close tabs on the monthly budget, to make early adjustments on spending if required. Allows the identification of whether costs are below, above, or aligned with projections, preventing potential losses.
Project Quality	• Create a checklist of required items. • Use resources from trusted suppliers. • Maintain regular communication with suppliers.	Employing reliable suppliers and maintaining regular communication reduces the risk of quality issues and delays.
Scope of the project	• Regular review of goal alignment with the Nike.	Ensures that both Nike and ReCon share the same project aims, and that the project plan fits the timeframe wanted by the Nike. Overall, this prevents misunderstanding, waste of time and resources.
Team Dynamic	• Conduct monthly surveys to collect feedback from team members about the work environment. • Regular team meeting encourages collaboration and allows the early detection of issues.	It is important to maintain a good team dynamic to increases the team's overall performance. Team members represent a share of the stakeholders; and are the core personnel behind the realization of the project, therefore their satisfaction is mandatory.
Client Satisfaction	• Frequent Communication between the technical project director and Nike. • Share summaries on the progress of the project to reassure Nike. • Respect Deadlines by closely following the Gantt Chart	It is fundamental to give Nike a positive experience, to build a trustful relationship and promote future projects with ReCon. This builds a positive reputation; and encourages Nike to recommend ReCon to other potential clients.
Changes of the project	• Make informed decisions after consulting with the relevant personnel. • Frequent communication between the managers and the rest of the team.	While some changes indicate proactive problem-solving, too many may suggest poor initial planning. Informed decisions help avoid repeated work and maintain quality throughout the project.

Format and Strategy	• Clearly divide tasks among the team. • Project managers should handle operational details, while directors focus on strategy. • Come to an agreement, on the format and frequency of stakeholder reports	Clearly dividing the work amongst the teams allows ReCon to obtain a descriptive and clear structure of the project. It also permits the decision of a sensible frequency and form of communication with stakeholders, to ensure their satisfaction.

Table 18 Key factors and their monitoring solutions

10.3 Project Monitoring Techniques

To manage these essential factors effectively, specific strategies are necessary. These techniques allow most of the project requirements to be controlled efficiently.

Project Monitoring Techniques	Justification
Gantt Chart	Gantt chart gives a clear visualization of project progress and achievements. It's useful for project team leaders to keep track of dates, end-product and deliverables, and key milestones that need to be met.
EV Chart	An Earned Value Chart allows to compare costs and progress estimations against the actual project developments. It is used to determine the overspending issues that can arise during the project.
Milestone slip chart	Milestone slip charts are used to show stakeholders the advancement of the project and which milestones have been accomplished. The use of this chart allows the modification of the schedule in case delays need to be addressed.
S – Curve	The S-curve is a graph representing the project's budget, progress, or workload spread out over time. It helps ensuring that the project aligns with the timeline and budget plans. By comparing the actual project curve to the estimated one regularly, tracking if the project is on track, ahead or delayed can be done. Frequent verifications allow enforcing solutions early on to avoid setbacks within the original project plan.
Refer To Business Cases	Ensures the project aligns with Nike's needs and the agreed timelines.
Continuously refer to key documents	To ensure a clear and efficient flow of the project, several key documents are created, including a budget cost table, risk P/I matrix, resource plan, PBS, WBS, along with a high level/milestone schedule. It is necessary to frequently come back to these documents to determine if the project is progressing as predicted.
Online and Face-to-Face Meetings	Regular meetings improve communication both internally (within the ReCon team) and externally (with Nike and other stakeholders). Enhanced communication reduces mistakes and facilitates progress tracking.

Table 19 Project Monitoring Techniques and Justifications

10.4 Responsibility, frequency and use of techniques

To implement these techniques effectively, it is crucial to assign responsibilities to qualified team members. Since these methods evaluate key factors, such as project adaptations and client communications, they should be managed by experienced project managers or directors.

Project managers should focus on operational details, while directors handle strategic decisions. Regular meetings between the two ensure agreement on formats and reporting schedules. For a project lasting 1.5 years, bi-weekly to monthly reports are recommended to identify and address underlying issues promptly. When a problem arises, rapid communication with relevant personnel is essential to prevent escalation.

10.5 Management of Project Material and Schedule Monitoring

Various tasks and subtasks overlap as the project progresses. For this reason, it is crucial to ensure that all team members and stakeholders obtain easy access to the necessary materials which are relevant to them (e.g. the Gantt Chart for the project managers or the Responsibility matrix for the ReCon team).

To allow an easy and clear access to material, and ensure nothing is lost along the way, material should be digitalized. This is done through documents (PDF, Excel Sheets, Gantt Project Files) which are saved in shared folders among the relevant parties.

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Appendix 1: Meeting Minutes

1) **Meeting 1: October 22**

1 hour zoom meeting, the entire team participated, and we worked on the first 2 sections together

2) **Meeting 2: October 18**

1.5 hours in person meeting, entire team was here, and we worked on more.

3) **Meeting 3: November 3**

1.5 hours meeting on teams, everyone joined we finalized PBS and worked on structuring previous parts, designated each person to do a specific task and organized another meeting

4) **Meeting 4: November 5**

Meeting from 12-2, we began working on the Gantt chart, and finished all stages before including WBS and PBS

5) **Meeting 5: November 9**

Meeting for 1.5h, we split the tasks such that one member focuses on costing, one on risk management and stakeholder management, one on resource plan and one on Gantt chart and schedule.

6) **Meeting 6: November 15**

Meeting of 1.5h to discuss the progress of our work. Started combining information together (e.g. resources + budget + schedule)

7) **Meeting 7: November 16**

Meeting of 2h: writing discussions where relevant, checking each other's work, adding information where relevant

8) **Meeting 8: November 17**

Meeting of 2h and message exchanges throughout the day to finalize the project. This included labeling tables, adding references, formatting, reviewing grammar and spelling, etc.

Note: all team members were present in every meeting.